

DATA 298B MSDA Project II

Final Project Report Requirements

1. Introduction

1.1 Project Background and Executive Summary

Project background, needs and importance, targeted project problem, motivations and goals. Planned project approaches and method. Expected project contributions and applications.

1.2 Project Requirements

Functional and AI-powered feature requirements which can be tested and measurable; data requirements.

1.3 Project Deliverables

Deliverables including reports, prototypes, development applications, and/or production applications.

1.4 Technology and Solution Survey

Survey of current technologies and solutions that could meet the project requirements. Summary and classifications of features and applications. Comparison of solutions including approaches, algorithms and models.

1.5 Literature Survey of Existing Research

Literature survey including summary and classification of research papers with justifications and contributions. Comparison among relevant research papers.

2. Data and Project Management Plan

2.1 Data Management Plan

Data collection approaches, management methods, storage methods, and usage mechanisms.

2.2 Project Development Methodology

Data analytics with intelligent system development cycle; planned development processes and activities.

2.3 Project Organization Plan

Work breakdown structure presenting the hierarchical and incremental decomposition of the project into phases, deliverables and work packages.

2.4 Project Resource Requirements and Plan

Required hardware, software, tools and licenses including specifications, costs and justification.

2.5 Project Schedule

Gantt Chart presenting project schedule with tasks, timeline, responsible team members, and the status of deliverables. PERT Chart performing project analysis with individual tasks and dependencies.

3 Data Engineering

3.1 Data Process

Decide the approaches and steps of deriving raw, training, validation and test datasets in order to enable the models to meet the project requirements.

3.2 Data Collection

Define the sources, parameters and quantity of raw datasets; collect necessary and sufficient raw datasets; present samples from raw datasets.

3.3 Data Pre-processing

Pre-process collected raw data with cleaning and validation tools; present samples from pre-processed datasets.

3.4 Data Transformation

Transform pre-processed datasets to desired formats with tools and scripts; present samples from transformed datasets.

3.5 Data Preparation

Prepare training, validation and test datasets from transformed datasets; present samples from training, validation and test datasets.

3.6 Data Statistics

Summarize the results of progressive results for including deriving raw, pre-processed, transformed and prepared datasets; statistically present the results in visualization formats.

3.7. Data Analytics Results

Present diverse data analytics results using diverse big data visualization formats, for example, map-based data analytics images, big data analytics diagrams.

4 Model Development

4.1 Model Proposals

Specify the applied, deployed, improved, proposed and/or ensembled models to each of the targeted problems in terms of concepts, inputs/outputs, features, model architectures, algorithms, etc.

4.2 Model Supports

Describe the platform, framework, environment and technologies supporting the development and execution of each model; provide diagrams of architecture, components, data flows, etc.

4.3 Model Comparison and Justification

For each targeted problem, compare the final selected and deployed models regarding the intelligent solutions, including strengths and targeted problems, approaches, data types, limitations; provide justification for each model.

4.4 Model Evaluation Methods

Present evaluation methods and metrics for each model, e.g., accuracy, loss, ROC/AOC, MSRE, etc. Specify the evaluation methods and metrics for each target problem and solution.

4.5 Model Validation and Evaluation Results

Present and compare detailed machine learning results based on selected model evaluation methods; present the solution to each targeted problem in terms of validated results, including accuracy, loss, etc. Include original images/data, result images/data, and validated images/data with detected/classified objects.

5. Data Analytics and Intelligent System

5.1 System Requirements Analysis

Describe system boundary, actors and use cases; describe high-level data analytics and machine learning functions and capabilities.

5.2 System Design

Present system architecture and infrastructure with AI-powered function components, system user groups, system inputs/outputs, and connectivity; present system data management and data repository design; present system user interface design, terms of system mockup diagram and dashboard UI templates.

5.3 Intelligent Solution

Present the developed AI and machine learning solutions for each targeted problem, including integrated solutions, ensembled, developed and applied machine learning models; describe required project input datasets, expected outputs, supporting system contexts, and solution APIs.

5.4 System Supporting Environment

Present the information and features of system supporting environment, including technologies, platforms, frameworks, etc.

6. System Evaluation and Visualization

6.1 Analysis of Model Execution and Evaluation Results

Evaluate the model output with tagged/labelled targets; describe the methodology of measuring accuracy/loss, precision/recall/F-score, or AUC, confusion metrics, etc.

6.2 Achievements and Constraints

Describe the achievements of solving the target problem(s) and the constraints have been encountered.

6.3 System Quality Evaluation of Model Functions and Performance

Evaluate the correctness of the model and the run-time performance of meeting system response time targets.

6.4 System Visualization

Apply visualization methodologies to present project data, analysis results, and machine learning outcomes, e.g., data analytics outcomes and map-based UI with different classification results.

7. Conclusion

7.1 Summary

Explain what the research has achieved; revisit key points in each section and summary of major findings, and implications for the field if any.

7.2 Benefits and Shortcoming

Discuss benefits and shortcoming of the solution presented.

7.3 Potential System and Model Applications

Discuss potential system and model applications

7.4 Experience and Lessons Learned

Discuss and summarize the experience and lessons learned from this project

7.5 Recommendations for Future Work

Provide recommendations for future project works and extensions.

7.6 Contributions and Impacts on Society

Describe the ways the project can contribute to the cultural, economic, educational and social well-being in diverse and multicultural local, national and global contexts.

References

List all references with proper citations using APA or IEEE format.

Appendices

Appendix A – System Testing

Present the test results of required use cases in terms of a sequence of GUI screens for each required use case.

Appendix B – Project Data Source and Management Store

Provide project data source information, e.g., training data, test data, etc. A Google Drive link will be provided to each group. Please upload of all created training, test data, and so on. Provide links of any pre-trained data.

Appendix C – Project Program Source Library, Presentation, and Demonstration

Upload project program artifacts, program source codes, PPTs, and demo videos to the Google Drive Link provided.